

EIDOS

ARC-AGI-3 grid world

abstract · turn-based

EXPANSE

NAO humanoid robot

physical · continuous

μ-GIL 2.0

A Spatial Abstraction Learner

Two worlds. One psyche. Nothing it did not earn.

A parsimonious instance of the Piagetian Modeler cognitive architecture,
built to navigate the ARC-AGI-3 spatial reasoning corpus.

Michael S. P. Miller · SubThought Corporation · BICA-26, September 2026

The Piagetian Modeler

The architecture both systems implement. Constructivist neurosymbolic: mental models built solely through perception.

Perception is active and constructive

1

Sensory input passes through detectors into a variable-length TRAIT VECTOR, compared against stored cases. Miss → assimilate: mint a reifier, store a case, the ontology grows. Hit → accommodate: reinforce, spread activation. A categorical decision — no loss function, no gradient.

The world model is acquired, not given

2

Every action produces proprioceptive feedback, compared against the prediction that preceded it. Confirmed reinforces; failed triggers compensation — the model is corrected at exactly the point of surprise.

One memory substrate, shared

3

Perception, coordination, reflection and consolidation read and write the same knowledge base. No separate training data, no separate inference model. Working memory IS long-term memory.

GIL — the reference implementation

~50 source files, over 20,000 lines of Premise. Four concurrent processes over one shared Totality.



Observation

3 patterns

Perceiver · Detectors · Matcher · Lexer · Storer · Activator · Imager (image schemas) ·
Locator / Mapper / Navigator



Coordination

10 solvers

Ameliorator (needs→objectives) · Deliberator · Reactor ·
Correlator/Regulator/Terminator · 136 reasoning operators, 7 strategies · Skemer ·
narrating



Reflection

8 patterns

motivation · simulation (daydreaming) · compensation · coping · discovery &
experimentation · imitation · play — developmentally gated



Consolidation

4 patterns

compressing (chunking) · automating (skill formation) · forgetting (decay pruning) ·
recycling (resource reuse)

THE TOTALITY

six memory areas, one substrate

Control	Registry · Actor · Mind
Ontological	Schemes :M :R :A :E
Perceptual	icons · geons · lexemes
Associative	Cases · Trait index
Activative	8 realities · Glue
Imaginative	Canvas · Taxis · Route

μ-GIL — the parsimonious instance

~30 source files: 13 mechanism theories, 2 psyches, totality, imagination, portal. ~8,000 lines of Premise, 1,200 of Python.

SHARES

- the perception pipeline, faithfully
- assimilation / accommodation as the only two paths
- Jaccard similarity matching at a configurable threshold
- content-addressed reifiers — identity survives restarts
- the Correlator grading every consequence
- an audit trace at every pipeline stage

REPLACES

- 7-level spatial hierarchy → empirical arrangement graph
- 10 concurrent solvers → single-threaded band auction
- 136 reasoning operators → learned transitions only
- 8 reflection patterns → simulation alone
- 4 consolidation patterns → a stale-cluster cull
- developmental gating, narration, theorizing → absent

INVENTS

- relational state layer — clusters, scale-normalised tokens
- anchor adoption — "this display can show what that plate shows"
- chain-walk — bounded BFS over the learned edge graph
- evidence confidence — majority-based edge recording
- the budget gate — learn the action budget from teleports
- question-the-map — accommodate the frontier ledger itself

The invented mechanisms are all responses to specific, diagnosed failures in live play — the kind a specification cannot contain.

Same architecture, two scales

No row is a different philosophy. Every μ -GIL entry is the GIL entry with what the task does not demand removed.

DIMENSION	GIL	μ -GIL
Source	~50 files, 20,000+ lines Premise	~30 files, ~8,000 Premise + 1,200 Python
Processes	4 concurrent, stigmergic	1 procedural loop, single-threaded
Memory areas	6, incl. Glue spatial layout	3 — case, scheme, activation
Realities	8 parallel layers	1 — everything is Observed
Spatial ontology	7 levels, designed a priori	4 layers, learned from transitions
Planning	10 solvers, 136 reasoning ops	~15 scoring bands, BFS depth ≤ 7
Forgetting	Amneator + 3 compressors	stale-cluster cull
Status	reference system, specified	runs live, wins level 1 reliably

Both answer the same question: how does a mind with no priors come to act competently in a world nobody described to it?

The Psyche — one contract, any world

A Psyche is the world interface — a sense organ. It mediates two flows and speaks exactly four tuples.

The four tuples



PERCEPT

inward

:Modality :Channel :Address :Data :Content :Moment



ATTEMPT

outward

:Action :Parameters :Token :By



RESULT

inward

:Action :Status :Reason :Moment :Token



URGE

inward

:Need :Source :Delta :Moment :Token

Interoception, proprioception, perception and recollection are four sequences over the same pipeline.

What varies, what does not

INVARIANT — the mind

- the tuple grammar and slot names
- Perceiver · Matcher · Lexer · Storer · Activator
- Scener · Simulator · Coordinator · Correlator
- every downstream stage works on the trait set alone

PER-DEVICE — the world

- :Channel — which sensory domain exists at all
- :Data — which parsable structures arrive
- :Content — the device's own slot vocabulary
- :Action — the device's actuation repertoire

World one — the Eidos

εἶδος, "to see." The device is the ARC-AGI-3 interactive reasoning benchmark. LS20, "Locksmith."

THE DEVICE

A 64×64 grid, 16 colours, 7 standardised actions, turn-based — the world changes only when the agent acts. No instructions are given: mechanics, goal and win condition must all be discovered. Scored by $RHAE = (\text{human actions} \div \text{agent actions})^2$.

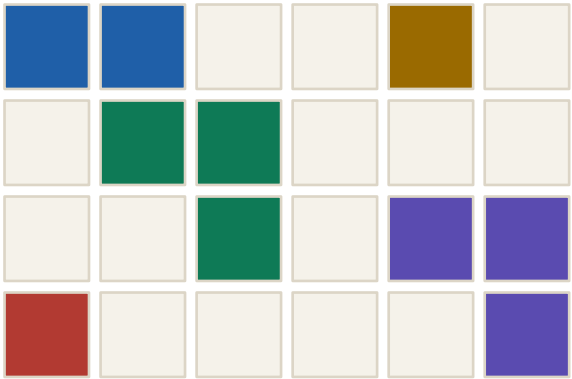
PERCEPTS • one channel: Grid

grid-state	full board, level, state, available actions
grid-delta	only what changed, keyed by causing action
game-event	win or game-over transition

ACTUATIONS • seven

RESET • ACTION1–4 (up/down/left/right) • ACTION5 (interact, select, rotate) • ACTION6 (click at x,y) • ACTION7 (undo)

WHAT THE MIND SEES



Two-pass connected-component labelling turns cells into regions; each region becomes one trait:

```
region 9:3x3:5@35,11
region 12:5x2:10@19,40
region-count 20    palette-size 8
```

~24 region traits per frame. Nothing in the mind knows what a lock, a key or a door is.

World two — the Expanse

Latin expansum, "spread out." The device is a NAO humanoid: 25 degrees of freedom, in a room.

THE DEVICE

SoftBank NAO over NAOqi or ROS 2. Two head cameras, a four-microphone array, torso IMU, sonar, bumpers, capacitive head sensors, 25 joints. Continuous, noisy — and it does not wait for the mind to finish thinking.

PERCEPTS • eight channels

Visual	camera-frame · depth-map · object-detected
Auditory	audio-sample · transcription · sound-event
Haptic	contact · bumper · collision
Proprioceptive	joint-state · imu-reading · battery-state
Spatial	odometry · pose · occupancy-grid
Communication	speech-heard · gesture-seen · touch-signal

ACTUATIONS • 35 in the spec

Locomotion	walk · turn · stop · stand · sit · navigate
Gaze	look · track
Manipulation	reach · grasp · release · lift · place
Gesture	point · wave · nod · shake-head
Communication	speak · emote · display · set-led
Motor	joint-move · pose-move · trajectory
System	e-stop · set-mode · capture-photo

HOMEOSTATIC URGES

power (battery < 15%) · thermal · balance (IMU tilt > 0.3 rad) · safety (collision) · connectivity

A body has needs a grid does not. Urges are how they reach the mind.

One interface, two devices

The tuple is identical. Every slot inside it is the device speaking — and detection is the only world-specific stage.

EIDOS

```
[PERCEPT :Modality Eidos      :Channel Grid      :Data grid-state
      :Content {:Grid {...4096 ints...} :State playing :LevelsCompleted 0
      :ActionsUsed 12 :HumanActions 8 :AvailableActions {...}]]
```

EXPANSE

```
[PERCEPT :Modality Expanse  :Channel Proprioceptive :Data joint-state
      :Content {:Joints [{:Name "LKneePitch" :Position 0.8 :Effort 3.1} ...]
      :Unit "rad"}]]
```

Matcher • Lexer • Storer • Scener • Activator • Simulator • Coordinator • Correlator operate on the trait set and nothing else. They cannot tell a grid from a knee. Add a world by adding a detector clause.

Spatial abstraction — designed vs empirical

The deepest divergence between the two systems, and the paper's most significant finding.

GIL — a designed ontology

The structural grammar is innate; experience fills it.



Spots bind objects to 3-D coordinates. Ten image schemas serve as spatial templates. The Locator builds bottom-up, the Mapper accumulates top-down, and a Navigator does BFS pathfinding over the graph.

Deductive: structure constructed from perceived relations by rule.

μ-GIL — an empirical ontology

No hierarchy, no image schemas, no prior spatial grammar.

1 Regions

two-pass connected components → colour, bbox, size, shape

2 Arrangements

controllable-colour whitelist → a multiset of "colour@x,y" keys ≈ GIL's Place

3 Edge graph

PairTried records (arrangement, action) → destination ≈ GIL's Map

4 Relational state

clusters with scale-normalised tokens; anchor adoption ≈ image schemas

Empirical: structure learned from observed transitions, one edge at a time. The chain-walk (BFS, depth ≤ 7) is μ-GIL's Navigator.

An independent parallel in vision theory

Blumberg's Neural Rendering describes the same epistemic structure μ -GIL's core loop implements.

THE FORMALISM • Blumberg (2026)

$$\hat{y} \text{ world } , \hat{y} \text{ body } = F (s \text{ world } , s \text{ body } , a) \quad (\text{i})$$

a forward model predicts world and body consequences

$$\delta = \text{encode}(y) - \hat{y} \quad (\text{ii})$$

the structured residual: predicted versus observed

$$s_{t+1} = G (s_t , \delta , a) \quad (\text{iii})$$

an empirically specified gate routes the residual

THE CORRESPONDENCE • in μ -GIL

F the TransitionScheme row — action + context $\rightarrow$ predicted moved / vanished / appeared

δ the Correlator's symmetric difference over predicted and observed arrangement keys

G the Correlator / Regulator / Terminator chain — reinforce, or overwrite at the point of surprise

The whitelist that separates the controllable piece from HUD churn is μ -GIL's separation of body state from world state.

NAPOT • receive $\rightarrow$ transform $\rightarrow$ re-express

Multiple partial projections converge at a receiver, the receiver integrates, and the result is re-expressed downstream. In μ -GIL the Matcher's Jaccard scan is this topology — traits converge at each Case, overlap is the integration, the 85% threshold is the quorum gate, the winner is re-expressed. So is the band auction (winner-take-all) and the chain-walk (value propagating back through a learned graph).

*The WHAT and WHERE
are present.
The WHEN is missing.*

Results — 45 live runs against the competition server

Up to 16 chained 8-turn sessions per run, the Totality saved and restored across every session boundary.

11

first live runs completed
zero levels

12

the run that first
completed level 1

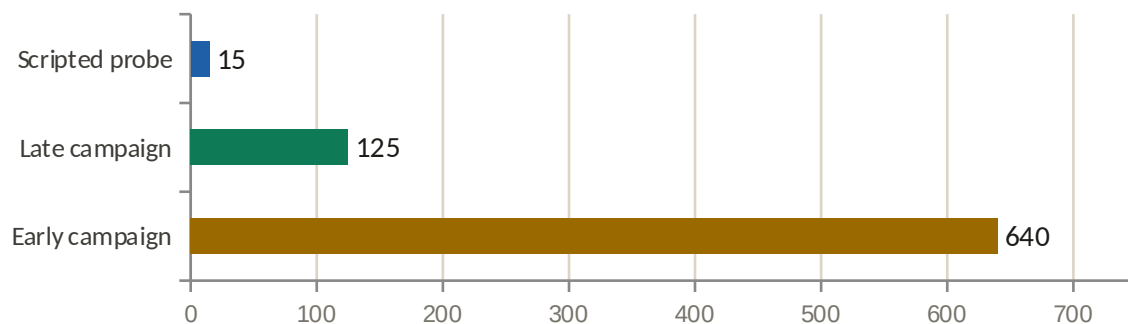
80%+

level-1 win rate from
run 14 onward

0

level-2 completions,
in any run

Cost per level-1 win



The gap between 125 and 15 is the price of building the abstractions rather than being given them.

The visible signature of learning

- Frontier burn-down**
plateaus at 30 in run 1; burns to 22 by run 7
- Assimilation ratio**
share of arrivals at arrangements seen before — climbs faster
- Exploration rate**
drops to zero: the mind has learned enough structure to exploit it
- Model confirmation**
surprises front-loaded, decaying as the world model converges

The boundary it drew

Level 2 was never completed. This is the most useful result in the campaign.

THE ENERGY ECONOMY

~21 actions per life, then the piece teleports home and the display reverts

~19 actions for the winning route — spawn, to the cross, to the portal

5 muse probes tested it; the furthest reached the cross, none entered the portal

THE SHAPE OF THE WALL

The winning mechanism sat 59 walk-steps away, inside the budget, and the mind could not see it. One-step dreams all read known-and-burned; the chain-walk found no frontier to route to; question-the-map won by default and took two-thirds of the level-2 action budget re-verifying a map that was already fine.

The diagnosis

What is missing is not compute or budget. It is DIRECTION — a reason to commit to a forty-step excursion whose value no one-step dream and no bounded walk can see. Evidence-driven exploration has a horizon, and forty-five runs found exactly where it ends. The mind cannot yet tell an untrustworthy map from a frontier that is out of reach: same signature, opposite causes.

*Plateaus break on
representation,
never on iteration.*

μ-GIL 2.0 — what changes

Tabula rasa, no imported direction, and the horizon closed by machinery the mind builds for itself.

Out

The muse channel

Waypoint routes written from outside. It moved the piece further than anything else — and it is the one thing the mind could not have earned.

In

Self-minted Routes

GIL's Router and Navigator already specify it: breadth-first search over the mind's own observed edges, producing a Route with waypoints, a cursor and a destination.

In

A Critic before acting

Rehearse the imagined outcome against the objective; promote only what survives. A doomed carry is vetoed before a budget gate has to catch it.

In

Failure as a first-class monad

"That did not work" becomes an impediment, an enablement hypothesis, and an agenda item to discover what causes it.

Fix

Graded identity and temporal activation

An inverted trait index returning a score, not a threshold that collapses two places into one — and the WHEN that NAPOT correspondence is missing.

None of these is a new idea. All five are already written in GIL — the work is porting them into what runs.

The long-horizon problem

In 2022 the longest task a frontier model could finish alone was seconds of human-equivalent work. Today's agents sustain hours — and the doubling interval is itself shrinking.

H1

intra-context

Many coupled steps in one window. Reasoning that verifies and recovers before the chain derails.

H2

cross-context

The task outgrows the window. State compressed, externalised, checkpointed, faithfully resumed.

H3

cross-task

An open-ended stream with shifting goals. Reusable skills that accumulate over time.



Mid-run verification, natively

Inline judging is where research is ahead of product. μ -GIL cannot act without first dreaming the outcome and settling it confirmed or surprised.



Durable state, tested to destruction

Sessions end every eight turns by design. Save, wipe, reload, resume — the H2 problem faced 45 times on the public record.



A real reward signal

A live competition API grades every action, and every decision's basis rides to its reasoning field. No simulated environment, no verifier to reward-hack.

The honest frontier

A mind that earns its knowledge is slow to start — and impossible to fool about what it does not know.



Proved

A minimal Piagetian mind wins a real ARC-AGI-3 level reliably from cold, learning its spatial ontology from scratch, with every decision auditable.



Bounded

Evidence-driven exploration has a horizon. Forty-five live runs found it precisely, and never crossed it without imported direction.



Next

Close the horizon from inside — then make the same core carry a NAO body.

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Building Sentient Beings

Miller & Blumberg (2025) · zenodo.org/records/15522356

Building Minds with Patterns

Miller (2018) · available on Amazon

Coding Artificial Minds

Miller (2026) · forthcoming

μ-GIL: A Spatial Abstraction Learner

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